ADVERTISING CLICK-THROUGH RATE (CTR) PREDICTION

Using Deep Learning: Multi-Layer Perceptron and Deep Cross Network

PROJECT FILE

Bachelor's in Computer Science & Engineering with Specialization in AI & ML

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DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

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CERTIFICATE

This is to certify that the project report titled "Advertising CTR Prediction using Deep Learning " has been submitted by the following students as part of their coursework for the degree of B. Tech CSE(AI/ML) at K.R. Mangalam University.

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This work has been carried out under our supervision and has not been submitted anywhere else for any other degree or examination.

Name of Supervisor: Mr. Kunal Rai

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ACKNOWLEDGEMENT

"Enthusiasm is the foundation of all progress — with it there is accomplishment, and without it there are only excuses."

We would like to take a moment to sincerely thank everyone who helped us throughout this project. When we first started, it felt quite overwhelming — especially since we were working with tools and techniques that were new to us. But with the right guidance and support, we were able to complete it successfully.

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A big thank you to our friends and classmates who helped us think through problems and kept us motivated along the way.

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EXECUTIVE SUMMARY / ABSTRACT

This project focuses on predicting whether a user will click on an online advertisement, a task commonly known as Click-Through Rate (CTR) prediction. CTR prediction is an important problem in digital advertising because it helps advertisers improve targeting, optimize campaign performance, and increase revenue by showing relevant ads to the right users.

The project uses the Avazu CTR Prediction dataset and formulates the task as a binary classification problem where the target variable indicates whether a user clicked on an advertisement or not. A large subset of 10,000,000 rows was loaded from the training data, and extensive preprocessing was performed to convert the raw categorical features into model-ready numerical form.

Two deep learning models were implemented and compared: a Multi-Layer Perceptron (MLP) and a Deep Cross Network (DCN v1). Both models used embedding layers for high-cardinality categorical features, while the DCN additionally modeled explicit feature interactions through cross layers.

The models were trained using PyTorch with mixed precision training and DataParallel support on two Tesla T4 GPUs. Their performance was evaluated using AUC-ROC and Log Loss on validation and test sets. The final test results showed that the MLP slightly outperformed the DCN, with a test AUC of 0.7391 compared to 0.7360 for DCN, and a lower test Log Loss of 0.5795 compared to 0.5979.

Keywords: CTR Prediction, Deep Learning, Avazu Dataset, MLP, Deep Cross Network, Embeddings, Binary Classification.

6. Introduction

6.1 Background

Online advertising platforms serve millions of ads every day, and one of their key challenges is estimating the probability that a user will click on a particular advertisement. This probability, known as click-through rate, plays a major role in ad ranking, bidding, and personalization systems.

Traditional advertising systems often rely on manually engineered features and linear models, but modern CTR tasks involve massive sparse categorical data and complex feature interactions. Deep learning methods provide a more flexible way to learn useful patterns from such large-scale data.

This project explores deep learning approaches for CTR prediction using the Avazu dataset and compares a standard Multi-Layer Perceptron with a Deep Cross Network.

6.2 Problem Statement

Digital advertising systems require accurate prediction of user clicks to improve ad relevance and optimize revenue. However, CTR prediction is difficult because the data is highly sparse, heavily categorical, large in scale, and imbalanced, with click events occurring far less frequently than non-click events.

The objective of this project is to build and compare deep learning models that can predict click probability from user, device, app, and site-related features in the Avazu dataset.

6.3 Objectives

Primary objective:
To build an effective CTR prediction system using deep learning and compare the performance of MLP and Deep Cross Network architectures on a large-scale advertising dataset.

Specific objectives:

1. To obtain and prepare the Avazu CTR dataset for model training.
2. To perform exploratory analysis on click distribution and selected input variables.
3. To extract time-based features such as hour of day and day of week.
4. To encode categorical variables using label encoding.

7. Data Description & Technical Stack

7.1 Dataset Description

The project uses the Avazu CTR Prediction competition dataset. The notebook downloads the competition files, extracts the compressed training file, and loads a subset of 10,000,000 rows for analysis and model development.

The loaded dataframe has 24 columns and includes the target column click along with categorical fields related to banner position, site, app, device, and anonymous engineered fields such as C1, C14, C15, C16, C17, C18, C19, C20, and C21.

The overall CTR in the selected sample is 16.61 percent, which indicates class imbalance because the majority of impressions do not result in clicks.

7.2 Data Dictionary

Feature Group	Description
click	Target label, 1 if the ad was clicked and 0 otherwise.
siteid, sitedomain, sitecategory	Website-related identifiers for ad placement context.
appid, appdomain, appcategory	App-related identifiers where the ad was shown.
deviceid, deviceip, devicemodel	Device-level identifiers and hardware context.
devicetype, deviceconntype	Device type and connection category.
bannerpos	Ad banner position on the screen/page.
C1, C14 to C21	Anonymous engineered categorical features from the dataset.

hourofday, dayofweek	Derived temporal features extracted from the packed hour column.

7.3 Technical Stack

Purpose	Tool / Library
Programming Language	Python
Data Handling	Pandas, NumPy
Deep Learning	PyTorch
Model Evaluation	Scikit-learn
Visualization	Matplotlib, Seaborn
Experiment Platform	Kaggle Notebook / Colab-style GPU environment
Hardware	2 Tesla T4 GPUs with mixed precision training support

8. Exploratory Data Analysis

Before model training, exploratory analysis was performed to understand the dataset structure, class imbalance, and selected feature behavior. The notebook generated visualizations for click distribution, CTR by hour of day, and CTR by banner position.

The class distribution plot confirmed that non-click impressions dominate the dataset, while the CTR by hour plot showed that click behavior changes over different hours of the day. The CTR by banner position plot also suggested that ad placement affects click probability.

The notebook saved these visualizations as eda.png, showing that EDA was used not only for summary statistics but also for understanding how temporal and placement-related features may influence user response.

9. Feature Engineering & Preprocessing

The raw hour field in the dataset was transformed into two useful temporal features: hourofday and dayofweek. After deriving these fields, the original raw columns id, hour, and hod were removed from the feature set.

All input features were treated as categorical and encoded using LabelEncoder. This step converted string or mixed categorical values into integer indices suitable for embedding layers in PyTorch models.

The notebook reports very high cardinality in some features, including about 786,741 unique values for deviceid and about 2,129,662 unique values for deviceip, which justifies the use of learned embeddings rather than one-hot encoding.

The final feature set contained 23 input variables, and the data was split into training, validation, and test sets in an 80:10:10 ratio using stratified sampling. All three splits preserved the same CTR of 16.61 percent.

10. Model Development & Training

10.1 Embedding Configuration

Since the Avazu data is highly categorical, each feature was mapped to an embedding space. Embedding dimensions were selected dynamically based on vocabulary size, capped at 50 dimensions, and the total concatenated embedding input size was 622.

This embedding-based representation allowed the models to learn dense, trainable feature vectors for sparse high-cardinality categories.

10.2 MLP Model

The first model used in the project was a Multi-Layer Perceptron. After embedding lookup and concatenation, the input passed through a sequence of fully connected layers with Batch Normalization, ReLU activation, and Dropout, followed by a final linear output layer.

The MLP acts as a strong baseline for CTR prediction because it can learn non-linear decision boundaries from dense embedded features.

10.3 DCN Model

The second model was a Deep Cross Network (DCN v1). In addition to the deep network branch, DCN included cross layers that explicitly modeled bounded-degree feature interactions using the original embedded input.

10.4 Training Setup

Both models were trained using BCEWithLogitsLoss and the Adam optimizer with weight decay. A learning-rate scheduler reduced the learning rate when validation loss stopped improving, and gradient clipping was applied for stability.

10.5 Parameter Count

The MLP model had 147,420,023 trainable parameters, while the DCN model had 147,424,377 trainable parameters. This shows that both models were similar in scale, with DCN being only slightly larger due to the cross-network components.

11. Results & Discussion

11.1 Validation Behavior

During training, both models started with strong validation AUC around 0.78 in the first epoch, but validation loss and AUC declined in later epochs, indicating possible overfitting as training progressed.

For the MLP, the best validation AUC recorded in the final summary was 0.7856, while the DCN achieved a best validation AUC of 0.7844.

Model	Parameters	Test AUC	Test Log Loss	Best Val AUC
MLP	147,420,023	0.7391	0.5795	0.7856
DCN	147,424,377	0.7360	0.5979	0.7844

The notebook declared MLP as the winner with an AUC advantage of 0.0031 over DCN. In this experiment, the simpler MLP performed slightly better than DCN on both AUC and Log Loss.

11.3 Visual Evaluation

The notebook generated a results dashboard saved as ctrresults.png, containing loss curves, validation AUC curves, test metric bars, ROC curves, score distributions, and calibration plots.

These visualizations are valuable for your report because they show not only final numbers but also model behavior over training, discrimination quality, and probability calibration.

11.4 Interpretation

The result suggests that although DCN is designed to model explicit feature interactions, it did not outperform the MLP in this particular implementation and training setup. Possible reasons include hyperparameter sensitivity, embedding configuration, limited number of cross layers, or the fact that the MLP was already expressive enough for the selected feature representation.

12. References

You can use this reference list in your report:

1. Avazu CTR Prediction Dataset, Kaggle competition data.
2. PyTorch Documentation.
3. Scikit-learn Documentation.
4. Matplotlib Documentation.
5. Seaborn Documentation.
6. Wang et al., Deep & Cross Network for Ad Click Predictions, 2017, referenced in the notebook architecture description.
7. Sample project report format referred from attached Group-5_Final-Report.pdf.